Parameter-Level Conditional Computation Does Not Buy Quality on Phones: Three Negative Results for On-Device LLMs

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Abstract

Token generation on phones is bound by memory bandwidth, so the quality a phone can serve is set by how many parameter bytes it can afford to read per token. A tempting way around this limit is *conditional computation at the parameter level*: use a cheap signal about the request (its task) to keep only the parameters that request needs in RAM, or to load a small task-specific module. We test this idea three ways on an Apple M4 Max, each against a fair dense or shared baseline. (1) Expert usage in Qwen3-30B-A3B is genuinely task-local (a task-predicted resident set of 32 of 128 experts per layer serves 64% of decode-time expert reads, versus 49% for a task-agnostic set), but the remaining misses still cost about 210 MB of flash reads per token at a 7 GB budget. (2) Routing only among resident experts removes those reads but collapses generation quality long before phone budgets (GSM8K 94% $\rightarrow$ 10% at 7 GB of resident weights), while a dense 8B model in 4.6 GB scores 89%; seven training-free rescue variants do not help. (3) A dense 4B core with per-task LoRA modules chosen per request (by oracle or by an embedding router) is within noise of a single shared multi-task LoRA on all four evaluations, at equal per-request cost and at equal storage. Along the way we find that fine-tuning on dataset reference answers damages a strong instruction-tuned core on every task, while self-distilled data does not, and that sentence-embedding routing misroutes 27% of DROP questions. Per-request selection appears necessary only for state that a shared model cannot hold, such as an individual user's data. All code, data, per-item scores and generations are released.

1 Introduction

During autoregressive decoding, each generated token reads every active weight once, so throughput on a bandwidth-bound device is roughly

$$\text{tokens/s} \approx \frac{\text{memory bandwidth}}{\text{active bytes per token}}. \tag{1}$$

Phones have small RAM and modest bandwidth compared with servers, which caps both the size of model that fits and the speed at which it runs. Mixture-of-experts (MoE) models decouple total from active parameters, and adapters (LoRA [7]) add task capacity for a few megabytes. Both suggest the same design: *decide once per request which parameters matter, and keep only those resident.* A request-level decision is attractive on device because it yields a fixed parameter set, hence a static compute graph, for the whole request, and loads from flash only at request boundaries.

We ask whether this design buys quality inside a fixed RAM budget, and answer it with three experiments of increasing generosity to the idea:

- **Exp. 1** measures whether MoE expert usage is predictable from the task at all (it is).
- **Exp. 2** keeps only task-predicted experts resident and forbids routing to the rest, so no flash reads occur during decoding; it fails, and training-free repairs do not help.
- **Exp. 3** gives the idea its best shot: a dense core plus small *trained* task modules, compared against one shared module of equal per-request cost and of equal storage; the task modules never win beyond noise.

All three are negative results. We report them in full because each rules out a design that is natural to try, and because together they point to where request-level selection *is* necessary (Section 6).

2 Setup

All experiments ran on a MacBook Pro (M4 Max, 36 GB unified memory, macOS 27) with MLX 0.31.2 and `mlx-lm` 0.31.3, using 4-bit community conversions of Qwen3 models [13] in non-thinking mode with greedy decoding. Benchmarks: GSM8K [5] (exact numeric answer), HumanEval [4], MBPP [2] (training only), and DROP [6]. Every script, the per-item scores behind each number, and all raw generations are in the repository (https://github.com/wasifhaq434701-png/Cortex-mem).

3 Experiment 1: is expert usage task-local?

Method. Qwen3-30B-A3B has 48 MoE layers with 128 experts each and routes every token to 8 experts per layer; the 4-bit model is 17.2 GB, of which 0.9 GB is non-expert weights. We generated 128 greedy tokens for 25 hand-written prompts (5 each for code, math, personal notes, chat and writing) and recorded every expert selection. For a budget of K resident experts per layer we measure the decode-time *hit rate*: the fraction of expert selections served by the resident set. Resident sets are chosen, leave-one-out, as the request's own top-K (oracle, an upper bound), the top-K of the other requests of the same task (*task*, what a request-level predictor could do), or the top-K of all other requests (*global*, static pruning).

Table 1: Decode-time hit rate (%) of the resident expert set, all 25 prompts. The last column converts task-predicted misses into flash reads per generated token.

K/128	resident GB	oracle	task	global	random	misses (MB/token)
16	2.9	60.3	41.7	29.3	12.5	594
32	4.9	81.8	64.2	49.1	25.0	365
48	7.0	92.5	79.4	65.4	37.5	210
64	9.0	97.5	88.8	77.9	50.0	114

Result. Locality is real (Table 1): at $K{=}32$ the task-predicted set serves 64% of reads against 49% for a task-agnostic set and 25% at random. The task signal matters most for code (56% vs. 32%) and math (59% vs. 33%); for chat and personal notes the task and global sets are close (62 vs. 59, 64 vs. 57), consistent with shared "general language" experts. Prefill on these short prompts touches 50–65% of all experts. The problem is the misses: even at $K{=}48$, about 79 expert loads per token remain, roughly 210 MB of reads per generated token. At 20 tokens/s that is $\sim$4 GB/s of random flash reads, far beyond what flash-offloading systems budget [1, 18]. Loading misses on demand is not viable, which motivates Experiment 2.

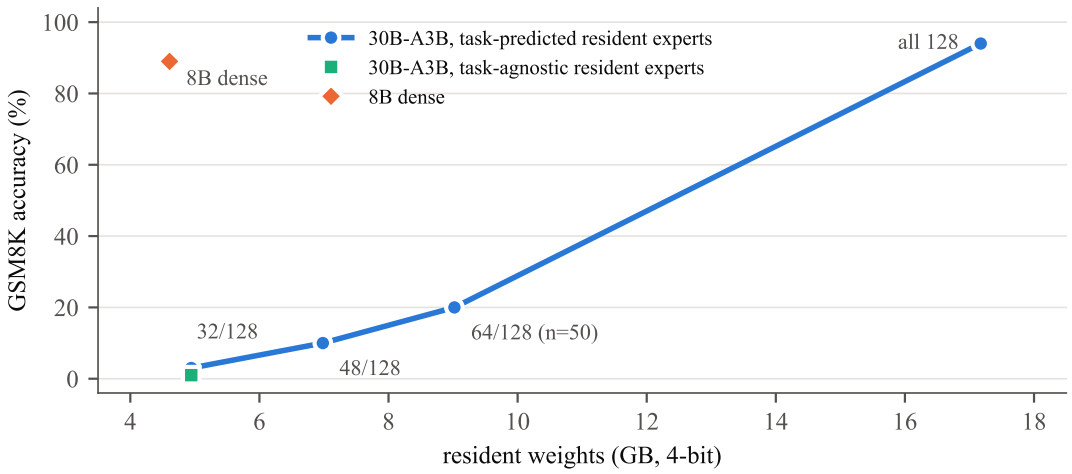


Figure 1: GSM8K accuracy against resident weight size. Masking Qwen3-30B-A3B's router to a task-predicted resident set collapses quality long before phone-scale budgets; a dense 8B model in 4.6 GB scores 89%. The 64/128 point uses the first 50 test questions.

4 Experiment 2: route only among resident experts

Method. We hard-mask the router so each token picks its top-8 experts *only* among the resident set ("reroute"), which eliminates flash reads during decoding. Resident sets are calibrated on the full model's routing over GSM8K train[:50] and HumanEval[:40], and evaluated on GSM8K test[:100] (accuracy) and on teacher-forced NLL per token of the canonical HumanEval[40:] solutions (124 problems). We also tried seven training-free rescue variants: larger budgets; "drop" (keep the router's original top-8, zero non-resident picks and renormalise); and adaptive per-layer budgets that give more slots to layers whose routing is more spread out, at equal total size.

Table 2: Constrained routing on Qwen3-30B-A3B. "task" / "global": resident set calibrated per task / once for all tasks. Rows below the rule use GSM8K test[:50].

configuration	resident GB	GSM8K (%)	code NLL/token
Qwen3-8B dense	4.6	89	0.671
Qwen3-30B-A3B, all experts	17.2	94	0.676
task set, K=48, reroute	7.0	10	0.788
task set, K=32, reroute	4.9	3	0.814
global set, K=32, reroute	4.9	1	1.354
task set, K=64, reroute	9.0	20	0.739
task set, K=64, drop	9.0	8	0.795
task set, K=48, drop	7.0	2	1.220
task set, K=32, drop	4.9	0	1.375
task set, adaptive 48/layer avg., drop	7.0	2	1.426
task set, adaptive 32/layer avg., reroute	4.9	4	2.030
task set, adaptive 32/layer avg., drop	4.9	2	1.968

Result. Quality collapses (Figure 1, Table 2). Keeping half the experts already drops GSM8K from 94% to 20%; at 7 GB it is 10%, while the dense 8B model scores 89% in 4.6 GB. We inspected the outputs: at K=64

generations stay coherent but wordier, and at $K{=}48$ they fall into repetition loops. Two observations matter beyond this model. First, teacher-forced NLL hides the cliff ($0.676 \rightarrow 0.788$ at $K{=}48$) while generation fails: small per-layer rerouting errors compound over 48 layers and over generated tokens. Second, task-calibrated sets beat the task-agnostic set at equal size (NLL 0.814 vs. 1.354), so the task signal is informative; it is simply not enough to replace training. "Drop" is worse than "reroute", and adaptive per-layer budgets hurt, suggesting that layers cannot be pruned independently. Post-hoc expert residency does not work without training; MoE models built for phones are instead trained for the device from the start [11], and task-level routing has succeeded when it is trained in [9].

5 Experiment 3: dense core plus per-task modules

Experiment 2 shows that parameters cannot be removed after training. The trained version of the same idea is a dense core plus small task modules, trained with the sparsity they run with, selected once per request and loaded from flash in megabytes rather than per token. This is the setting of LoRA libraries and routers such as LoraHub [8], MoLE [17], X-LoRA [3] and Arrow [12]. The question is whether choosing a module per request beats simply training one shared module.

Design. Core: Qwen3-4B (4-bit). Three tasks, each with training data disjoint from its evaluation: *math* (GSM8K train $\rightarrow$ GSM8K test[:250], accuracy); *code* (all 974 MBPP problems $\rightarrow$ HumanEval, 164 problems, pass@1 by executing the official tests in an isolated subprocess); and *notes*, an equal mix of DROP training questions and a synthetic personal-notes benchmark we release. Its items are short notes mixing bills, birthdays, medication and warranties, with questions whose answers must be computed (sums, date differences, expiry years), and the other facts act as distractors. Train and test use disjoint name, item and medication pools and disjoint phrasings of every fact and question, so test gains require transfer rather than template memorisation. Notes is evaluated on 200 held-out synthetic items (exact match) and 300 DROP validation questions (EM/F1, number and span answers), always reported separately.

Adapters: LoRA on every linear layer of all 36 blocks, rank 8 (66 MB as saved), batch 4, prompt-masked loss, maximum length 1024. Baselines: no adapter; *multi-r8*, one LoRA on the union of the three training sets with the same rank (equal per-request cost); and *multi-r24*, rank 24 (equal storage to all three task adapters). The shared adapters train for three times as many steps, so they see as many examples as the three task adapters combined. Because `mlx-lm`'s LoRA scale multiplies the update directly, we keep scale$\times$rank constant (scale 20 at rank 8, $20 \cdot 8/24$ at rank 24); rank 24 at scale 20 diverged within 40 steps. Every adapter is evaluated on every set, so the *oracle* (each item scored under its own task's adapter) and *routed* conditions follow from the same per-item scores without reloading models. The router is Cortex-style [14]: it embeds the request with all-MiniLM-L6-v2 [15] and picks the task whose centroid of 300 training prompts is nearest by cosine similarity.

Round 1: reference answers damage the core. We first trained on dataset reference answers (GSM8K solutions, MBPP code, synthetic rationales, DROP gold answers; 600 steps, learning rate 10^{-4}). Every adapter got worse on its own task except notes (Table 3). Inspecting the outputs shows why: GSM8K references and MBPP solutions are terser and sloppier than the 4B's own answers, so the adapters learned a worse style. They also collapsed off-task: the code adapter answers GSM8K questions with code (6%), and the math adapter writes prose instead of code (HumanEval 3.7%). This matches the distribution-gap explanation of fine-tuning damage in [19].

Round 2: self-distilled data (the reported run). Following [19], we regenerated every training set from the base model's own greedy answers and kept only correct ones: matching GSM8K answer (1809 of 2050

Table 3: Round 1 (reference-answer fine-tuning). Rows are adapters, columns evaluation sets (%). Diagonal entries in bold.

adapter	GSM8K	HumanEval	notes EM	DROP EM
none (base)	88.0	77.4	57.5	60.7
math	**63.2**	3.7	37.5	36.0
code	6.0	**32.3**	4.5	6.7
notes	15.6	0.0	**74.0**	**40.7**
multi-r8	57.2	37.2	74.5	48.7

kept), passing all MBPP tests (618 of 974), and exact match on notes and DROP (1048 of 2050); examples longer than 1024 tokens were dropped. We trained 300 steps per task adapter (900 for the shared adapters) at learning rate $2 \cdot 10^{-5}$. Uncertainty is a 95% paired bootstrap over evaluation items (10,000 resamples). The base model's DROP EM differs from round 1 by 0.6 points (61.3 vs. 60.7) because the two runs batched prompts differently, which changes a few numerically borderline greedy generations.

Table 4: Round 2 (self-distilled fine-tuning), % on each evaluation set (n = 250, 164, 200, 300).

configuration	GSM8K	HumanEval	notes EM	DROP EM	DROP F1
no adapter	88.0	77.4	57.5	61.3	69.1
math adapter	88.0	79.3	66.0	65.0	70.3
code adapter	88.8	76.2	58.0	60.0	67.2
notes adapter	88.8	78.0	69.0	66.3	72.1
oracle task adapter	88.0	76.2	69.0	66.3	72.1
routed task adapter	88.0	76.2	69.0	64.0	69.6
shared multi-r8 (equal cost)	86.4	80.5	67.0	66.0	71.6
shared multi-r24 (equal storage)	88.4	78.0	67.5	64.3	70.7

Result. Self-distillation removes the damage: no adapter is below the base on any set (Table 4). The oracle task adapters improve on the bare core only where it had headroom: notes $+11.5$ points [5.5, 17.5] and DROP $+5.0$ [0.3, 9.7]; math and code are unchanged, as training on answers the model already gets right teaches it little. The central comparison is null (Figure 2). Oracle task adapters versus the equal-cost shared adapter differ by $+1.6$, -4.3, $+2.0$ and $+0.3$ points, and versus the equal-storage shared adapter by -0.4, -1.8, $+1.5$ and $+2.0$; every interval contains zero. Much of the notes gain is not even task-specific: the *math* adapter alone improves notes by 8.5 points, plausibly through more reliable answer formatting. The router is perfect on GSM8K and notes and 99.4% on HumanEval, but routes only 73% of DROP questions to the notes adapter (66% in round 1), because number-heavy passages look like math to a sentence embedding. This costs 2.3 points of DROP EM relative to the oracle.

6 Discussion

What the three results rule out. With a cheap request-level signal, it is possible to predict which parameters a request uses (Exp. 1). But removing the rest after training destroys generation (Exp. 2), and training separate modules to select among buys nothing over training one shared module (Exp. 3). In Exp. 3 the shared adapter had no interference to avoid: once the training data matched the core's own distribution, no adapter hurt any other task. A conditional design only pays when what is selected cannot be shared.

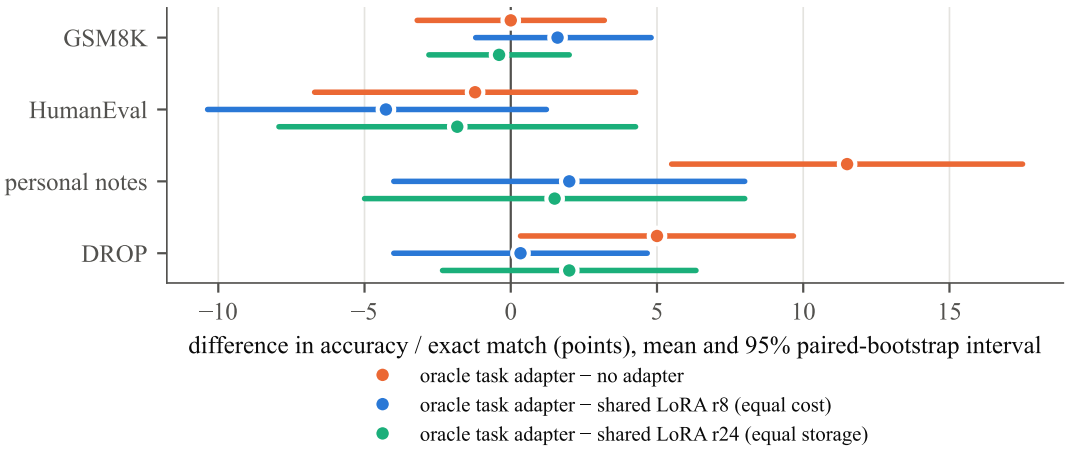


Figure 2: Paired differences with 95% bootstrap intervals. Task adapters beat the bare core only on personal notes and DROP, and are indistinguishable from a single shared LoRA everywhere.

Where selection is necessary. One class of state cannot be shared by construction: an individual user's data. A single model cannot contain every user's private history, so per-user state must be stored separately and selected per request. This argues for spending the device's conditional-computation budget on a user's memory rather than on model parameters: select which user-specific context or state to bring into a fixed RAM budget for a request, as context selection does for local retrieval [14]. We are evaluating this on long-term memory benchmarks such as LongMemEval [16] and LoCoMo [10], and measuring on a phone.

Limitations. All results use one core model per experiment, a single seed, and a laptop rather than a phone; on-device speed, energy and thermal behaviour are not measured here. Experiment 1 uses 25 hand-written prompts. Experiment 3 has three tasks, two of which the core already handles well; tasks where a 4B core is weak may show larger adapter gains, and with more tasks a single rank-8 adapter may saturate where a library would not. The personal-notes benchmark is synthetic and was built by us. Mixed-task requests, the expected weak point of request-level routing, were not tested; given the null result on single-task requests we do not expect them to reverse the conclusion.

7 Reproducibility

Code, training data for both rounds, per-item scores, every generation and every training log are in the repository, https://github.com/wasifhaq434701-png/Cortex-mem:

- Exp. 1: `experiments/expert_locality.py`
- Exp. 2: `experiments/routing_quality.py`
- Exp. 3: `experiments/task_adapters.py` (commands `distill`, `train`, `eval`, `report`) and `experiments/notes_synth.py`

Experiment 3 takes about 11 hours on an M4 Max; `report` recomputes every Exp. 3 number and interval from the committed per-item scores without a GPU. Adapter weights (about 700 MB) are available from the author.

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